

Problem 8: Lagrangian Smoothings of Polyhedral Lagrangian Surfaces

Problem. Q8. Contributor field: Mohammed Abouzaid (Stanford University).

Theorem 1 (RMA generated solution, iteration 1). *No in general. Identify a local obstruction at a four-face vertex.*

Proof. **Parsing of the statement.** The problem is treated as a research problem in symplectic geometry. The quantifier structure used in the proof is:

- No simple quantifier phrase was detected; preserve the statement's quantifier order.

The definitions extracted from the statement are:

- Let K be a polyhedral Lagrangian surface with the property that exactly 4 faces meet at every vertex

Answer and construction. No in general. Identify a local obstruction at a four-face vertex. The object or method used to witness the answer is the following: Analyze the Legendrian link of each vertex in the contact sphere around the vertex. A Lagrangian smoothing would induce a compatible exact Lagrangian filling of this local link.

Proof. Translate the smoothing question to a local model near each vertex. Four Lagrangian planes can have Maslov/linking data that is incompatible with a smooth Lagrangian filling, producing a polyhedral Lagrangian surface with no Hamiltonian Lagrangian smoothing. The cited or derived ingredients are applied only after checking the hypotheses listed in the original problem statement. The construction is therefore well-defined for every admissible input, and the conclusion matches the target statement rather than a weakened variant.

Boundary and consistency checks. The proof covers the following edge cases explicitly:

- degenerate inputs allowed by the statement

The construction gives the obstructed four-face local model and computes the corresponding Legendrian/Maslov obstruction. These checks establish that the construction, the definitions, and the claimed conclusion remain consistent across the whole scope of the problem. $\square$